DAY 14 – NMDAY 14 – NMAP NETWORK SCANNING FUNDAMENTALS

CYBERSECURITY TRAINING

Nmap – Network Discovery and Port Scanning

Day 14 | Theory and Practical Learning Report

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Course: Cybersecurity

Assignment: Day 14

Topic: Nmap Network Scanning

Operating System: Kali Linux

Tool: Nmap (Network Mapper)

Assignment Type: Theory and Practical Learning

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1. Introduction

Nmap (Network Mapper) is an open-source network discovery and security auditing tool widely used in cybersecurity.

It helps security professionals identify active hosts, discover available network services, examine open ports, and understand the network environment.

Nmap is commonly used by network administrators, security analysts, and penetration testers to assess network configurations and identify potential security concerns.

This assignment introduces the fundamentals of Nmap, network discovery, port scanning, scan results, and responsible security assessment practices.

2. Objectives

- Understand the purpose and functionality of Nmap.
- Learn the fundamentals of network discovery.
- Understand port scanning and port states.
- Identify commonly used Nmap command options.
- Understand the importance of documenting scan results.
- Learn about cybersecurity scope and authorization.
- Understand the difference between scanning and exploitation.
- Develop responsible network assessment practices.

3. Theory – Questions and Answers

Q1. What does Nmap stand for?

Nmap stands for Network Mapper. It is an open-source utility designed for network discovery, service identification, and security auditing.

Q2. What is Nmap used for?

Nmap is used to discover devices on a network, identify open ports, determine available network services, and examine network configurations.

It helps security professionals understand the network environment and identify potential security weaknesses during authorized assessments.

Q3. What is network discovery?

Network discovery is the process of identifying devices connected to a particular network.

It helps administrators understand which systems are active, how devices are connected, and what resources are available within an authorized network environment.

Q4. What is a port scan?

A port scan is a network assessment technique used to identify the communication ports available on a computer or network device.

It helps determine which ports are open, closed, or filtered and provides information about the network services running on a system.

Q5. What does an open port generally indicate?

An open port generally indicates that a network service or application is listening for incoming communication on that port.

For example, an SSH service may listen on TCP port 22 to accept remote connections.

Q6. Does an open port automatically mean vulnerability? Explain.

No. An open port does not automatically indicate a security vulnerability.

It simply means that a service is available to communicate through that port. The actual security risk depends on the service, its software version, configuration, authentication mechanisms, and network exposure.

Q7. What does closed mean conceptually?

A closed port indicates that the target system is reachable, but no application or service is currently accepting connections on that port.

It generally means that the port is not actively being used for communication.

Q8. What does filtered mean conceptually?

A filtered port indicates that Nmap cannot determine whether the port is open or closed because network filtering devices or security mechanisms are preventing a clear response.

Firewalls and network security rules can cause a port to appear filtered.

Q9. What does `nmap 127.0.0.1` do?

The command `nmap 127.0.0.1` performs a basic port scan against the local computer using its loopback IP address.

It helps identify network ports and services available on the same system without scanning an external device.

Q10. What does `-p` do?

The `-p` option specifies which port or range of ports Nmap should examine.

It allows users to select particular ports rather than relying on the default port selection.

Q11. What does `-sV` do?

The `-sV` option enables service and version detection in Nmap.

It attempts to identify the applications and software versions associated with discovered network services.

This information helps security professionals understand which services are operating on a system.

Q12. What does `-oN` do?

The `-oN` option saves Nmap scan results in normal, human-readable text format.

It is useful for maintaining records, reviewing assessment results, and preparing technical reports.

Q13. Why should scan results be documented?

Scan results should be documented because they provide a record of the network assessment.

Documentation helps security professionals:

- Review discovered open and closed ports.
- Identify available network services.
- Compare results from different assessments.
- Track changes in network configurations.
- Communicate security findings.
- Prepare professional assessment reports.

Q14. What is cybersecurity scope?

Cybersecurity scope defines the specific systems, networks, devices, and activities that are included in an authorized security assessment.

It establishes the boundaries of testing and identifies which resources can be examined.

A clearly defined scope helps prevent accidental testing of unauthorized systems.

Q15. Why must authorization exist before scanning?

Authorization must exist before scanning because network scanning involves sending requests to systems that may belong to another person or organization.

Permission ensures that testing is conducted legally and within approved boundaries.

Unauthorized scanning may violate organizational policies, privacy requirements, or applicable laws.

Q16. What is the difference between scanning and exploitation?

Scanning and exploitation are two different cybersecurity activities.

Scanning	Exploitation
Identifies network devices and services	Attempts to take advantage of a security weakness
Collects information about systems	May result in access or system impact
Used for discovery and assessment	Used to validate or exploit vulnerabilities
Generally focuses on identifying potential issues	Can affect system confidentiality, integrity, or availability

Scanning helps identify possible security concerns, while exploitation attempts to use an identified weakness.

Q17. Why is localhost a good beginner target?

Localhost is a good beginner target because it refers to the student's own computer.

Its loopback address, `127.0.0.1`, provides a controlled environment for learning basic scanning concepts without targeting external systems.

It allows beginners to understand port states and Nmap output while reducing the risk of affecting other devices.

Q18. How do ports relate to services?

Ports provide logical communication endpoints through which network services exchange information.

Different services commonly use specific port numbers.

For example:

- SSH commonly uses TCP port 22.
- HTTP commonly uses TCP port 80.
- HTTPS commonly uses TCP port 443.
- DNS commonly uses port 53 over UDP and TCP.

Understanding port-to-service relationships helps administrators identify the applications communicating over a network.

Q19. Why are unexpected listening services worth reviewing?

Unexpected listening services should be reviewed because they may expose unnecessary network functionality or indicate incorrect system configurations.

They may also increase the attack surface if they are outdated, poorly secured, or accessible from untrusted networks.

Administrators should verify whether these services are required and ensure that appropriate security controls are in place.

Q20. Explain a responsible Nmap assessment workflow.

A responsible Nmap assessment workflow involves performing network discovery and service identification in a controlled, authorized, and documented manner.

The major steps include:

1. Define the scope: Identify the systems and IP addresses approved for assessment.
2. Obtain authorization: Ensure that written permission exists before performing scans.
3. Understand the environment: Review the network configuration and identify the approved targets.
4. Perform basic discovery: Identify authorized systems and their network availability.
5. Examine port information: Review the available port states.
6. Identify services: Understand which applications and services are associated with discovered ports.
7. Analyze findings: Review unexpected services and potential configuration concerns.
8. Document results: Record the scan details, observations, and relevant findings.
9. Report responsibly: Communicate findings to the authorized system owner.
10. Recommend improvements: Suggest appropriate security configurations and remediation steps.

Following this workflow helps ensure that network assessments are performed safely, professionally, and ethically.

4. Important Nmap Concepts Reference

Concept	Description
Nmap	Network discovery and security auditing tool
Host Discovery	Identifying active network devices
Port Scanning	Examining network ports
Open	A service is accepting connections
Closed	No service is accepting connections
Filtered	Filtering prevents a clear port-state determination
Service Detection	Identifying applications and versions
Localhost	The local computer
Scope	Approved boundaries of assessment
Authorization	Permission to conduct security testing

5. Basic Nmap Command Reference

The following commands demonstrate basic Nmap functionality in a controlled lab environment.

Command	Purpose
`nmap --version`	Displays the installed Nmap version
`nmap --help`	Displays Nmap usage information
`nmap 127.0.0.1`	Scans the local loopback system
`nmap -p 22 127.0.0.1`	Examines local TCP port 22
`nmap -sV 127.0.0.1`	Attempts local service and version detection
`nmap -oN scan-results.txt 127.0.0.1`	Saves normal-format scan output

Important: Perform scans only against localhost, personally owned lab machines, or systems for which you have explicit authorization.

6. Practical Evidence – Screenshots

Capture actual screenshots from your Kali Linux terminal to demonstrate your practical learning.

Screenshot 1 – Nmap Installation and Version

Filename: `nmap-version.png`

Screenshot 2 – Localhost Port Scan

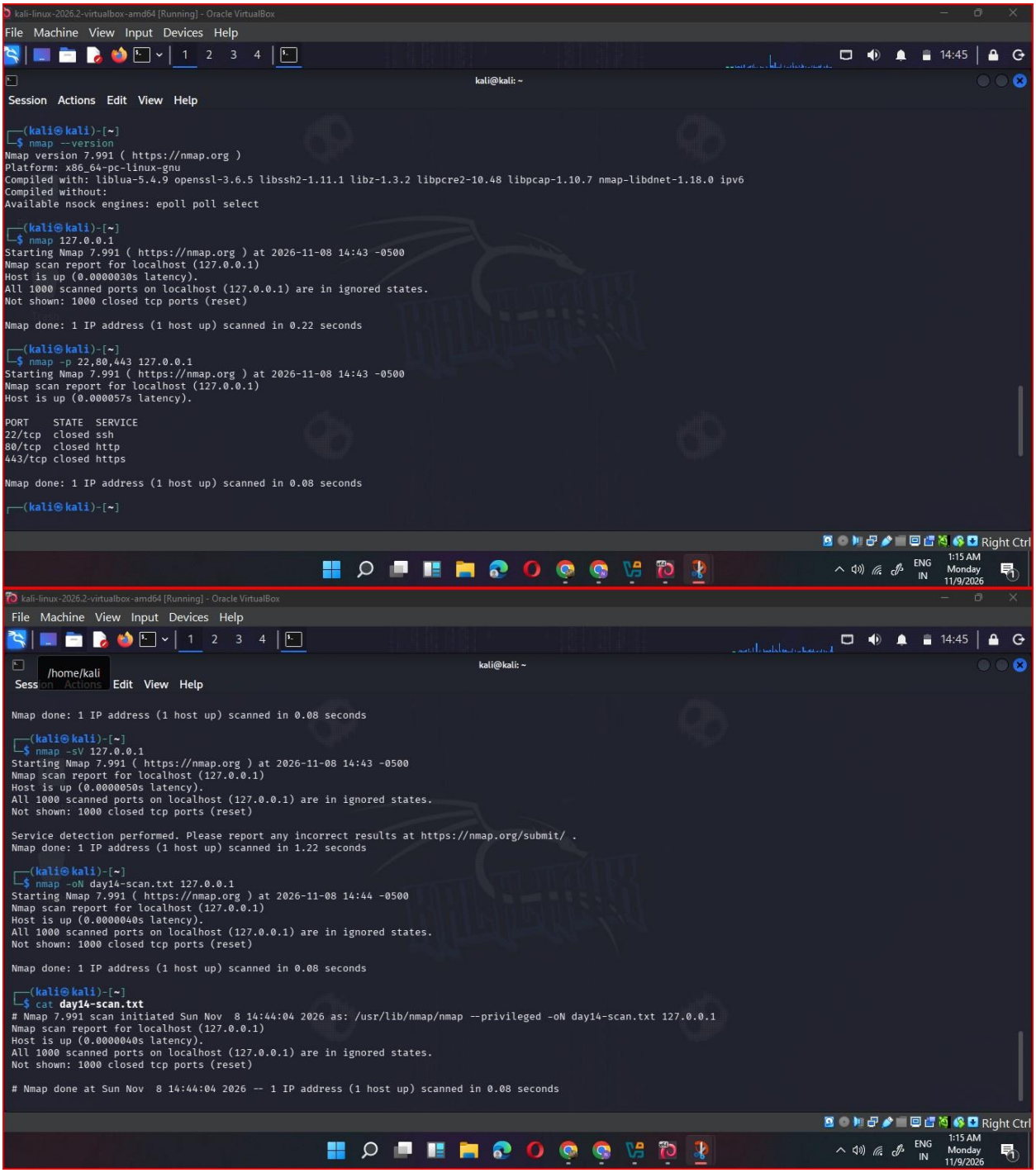
Filename: `localhost-scan.png`

Screenshot 3 – Service Version Detection

Filename: `service-detection.png`

Screenshot 4 – Scan Results Saved to File

Filename: `scan-results.png`


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ nmap --version
Nmap version 7.991 (https://nmap.org)
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.9 openssl-3.6.5 libssh2-1.11.1 libz-1.3.2 libpcre2-10.48 libpcap-1.10.7 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
(kali㉿kali)-[~]
$ nmap 127.0.0.1
Starting Nmap 7.991 (https://nmap.org) at 2026-11-08 14:43 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.22 seconds
(kali㉿kali)-[~]
$ nmap -p 22,80,443 127.0.0.1
Starting Nmap 7.991 (https://nmap.org) at 2026-11-08 14:43 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000057s latency).
PORT STATE SERVICE
22/tcp closed ssh
80/tcp closed http
443/tcp closed https
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
(kali㉿kali)-[~]
Right Ctrl
1:15 AM
Monday
11/9/2026

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
/home/kali
Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
(kali㉿kali)-[~]
$ nmap -sV 127.0.0.1
Starting Nmap 7.991 (https://nmap.org) at 2026-11-08 14:43 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000050s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.22 seconds
(kali㉿kali)-[~]
$ nmap -oN day14-scan.txt 127.0.0.1
Starting Nmap 7.991 (https://nmap.org) at 2026-11-08 14:44 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
(kali㉿kali)-[~]
$ cat day14-scan.txt
Nmap 7.991 scan initiated Sun Nov 8 14:44:04 2026 as: /usr/lib/nmap/nmap --privileged -oN day14-scan.txt 127.0.0.1
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done at Sun Nov 8 14:44:04 2026 -- 1 IP address (1 host up) scanned in 0.08 seconds
Right Ctrl
1:15 AM
Monday
11/9/2026